

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

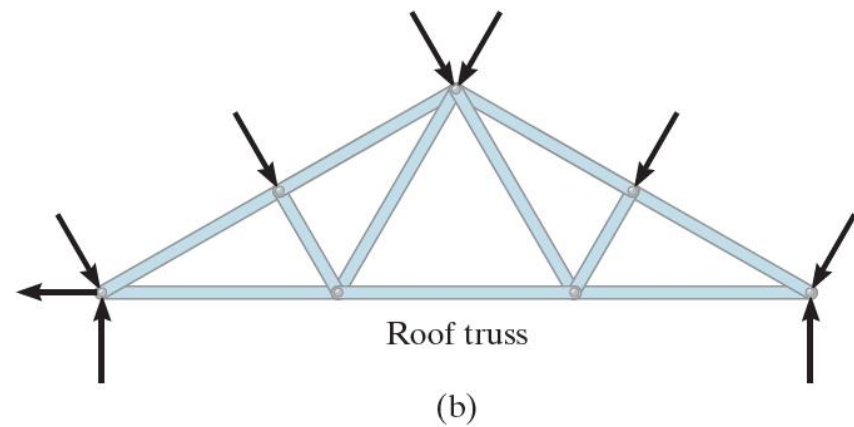
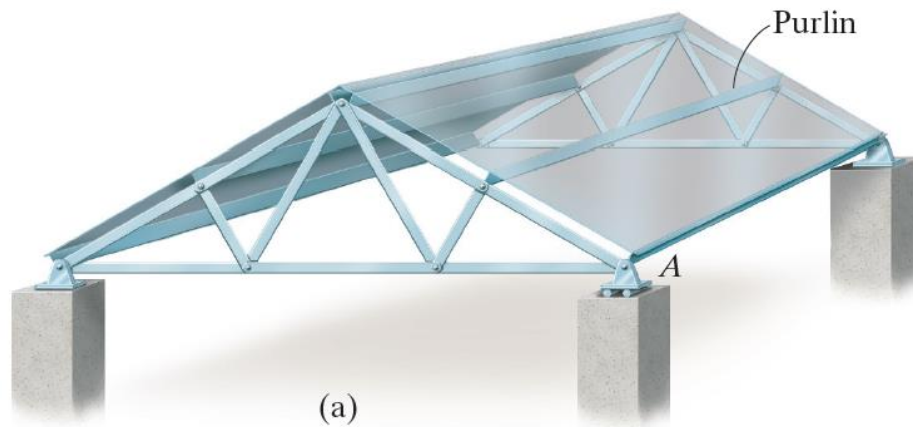
Lecture 9:

Structural Analysis

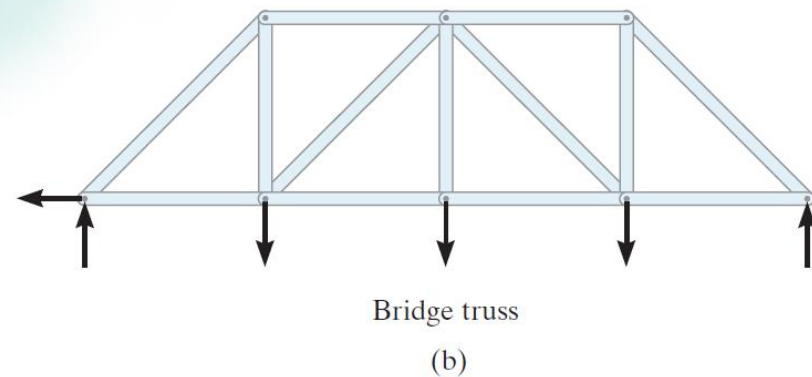
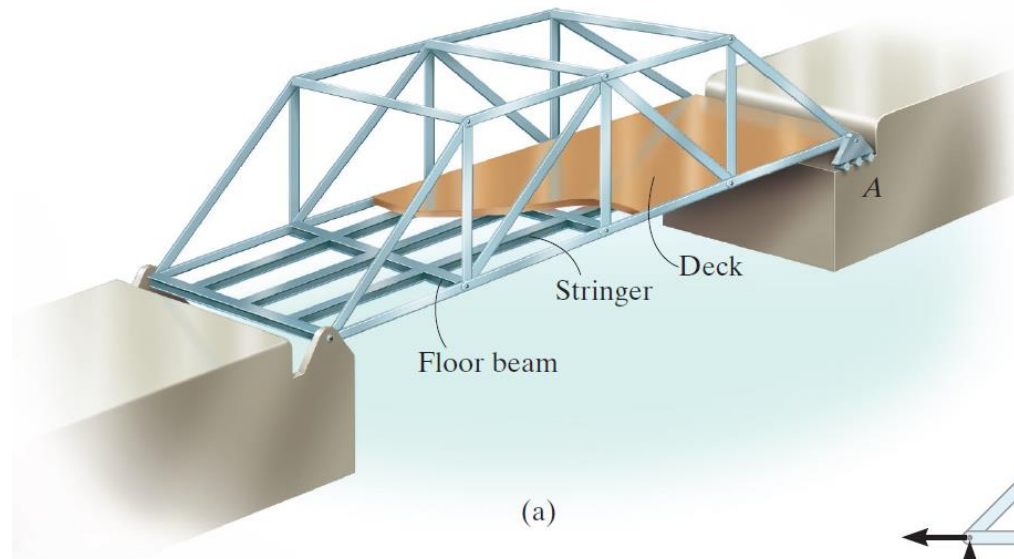
By Saeed Lotfan

Simple Trusses

A **truss** is a structure composed of slender members joined together at their end points. The members commonly used in construction consist of **wooden struts** or **metal bars**.



Planar Trusses

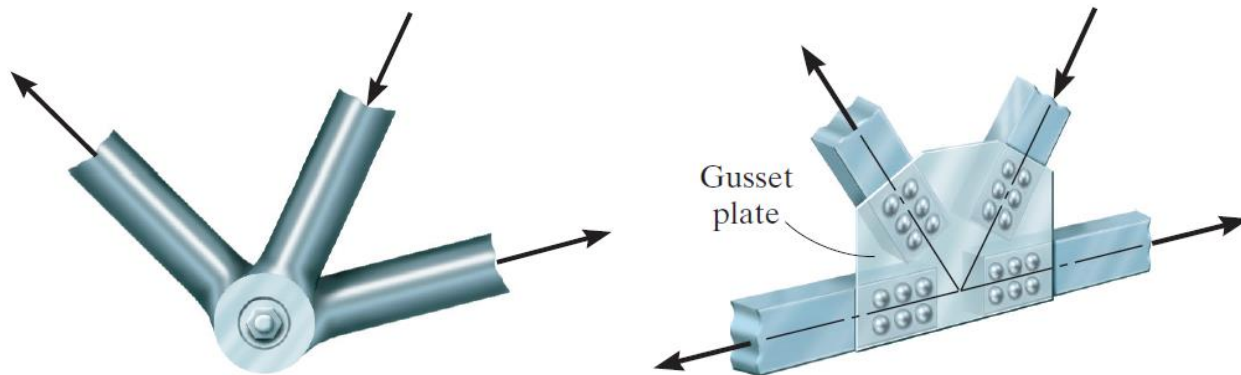


- The loadings can be investigated based on two-dimensional method.
- The type of support should allow freedom for expansion or contraction of the members due to a change in temperature or application of loads.

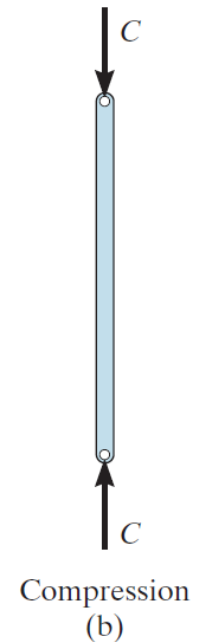
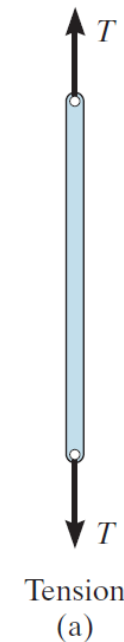
Assumptions for Design

To design both the members and the connections of a truss:

- **All loadings are applied at the joints.**
- **The members are joined together by smooth pins.**



Because of these two assumptions, each truss member will act as a two-force member.

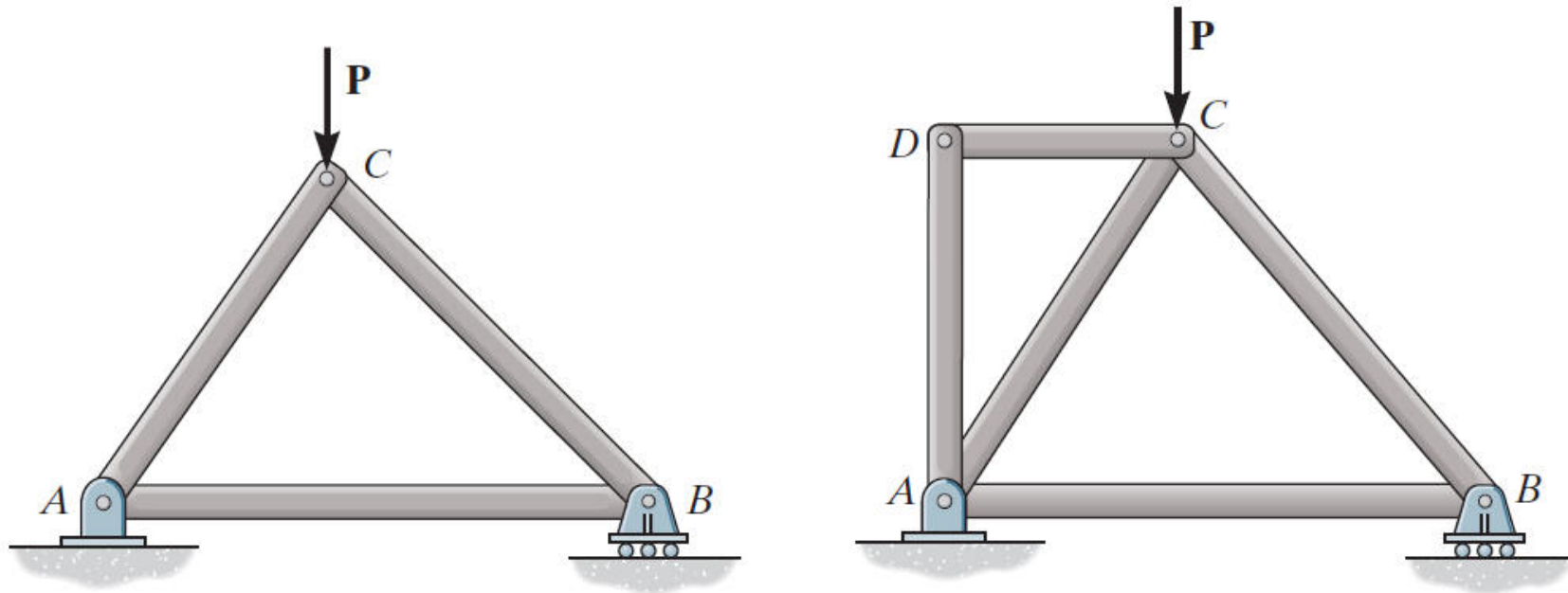




The use of metal gusset plates in the construction of these Warren trusses is clearly evident.

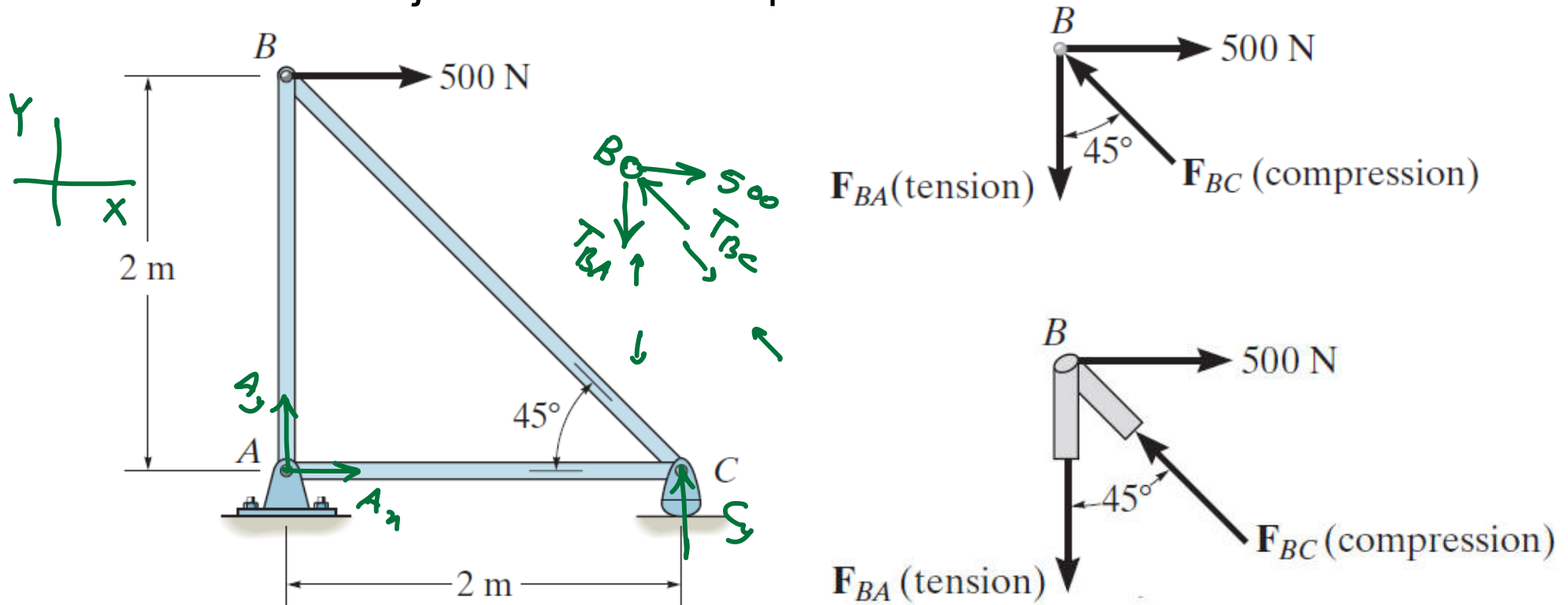
Simple Truss

If three members are pin connected at their ends, they form a **triangular truss** that will be rigid. Attaching two more members and connecting them to a new joint D forms a larger truss. This procedure can be repeated as many times as desired to form an even larger truss, and by doing this one forms a simple truss.



Methods of Analysis: The Method of Joints

This method is based on the fact that if the entire truss is in equilibrium, then each of its joints is also in equilibrium.



When using the method of joints, always start at a joint having *at least one known force and at most two unknown forces*.

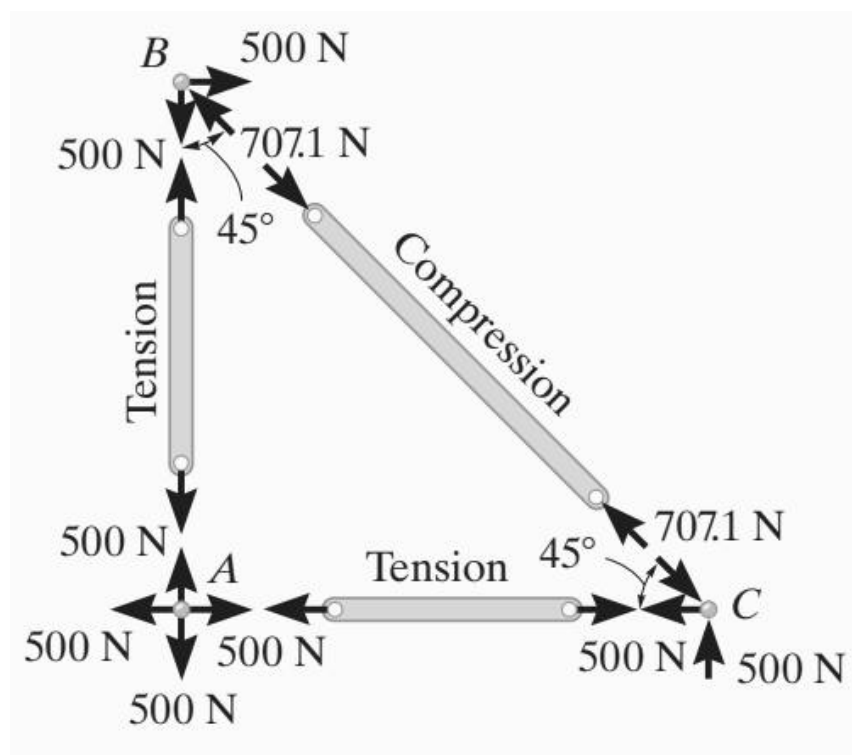
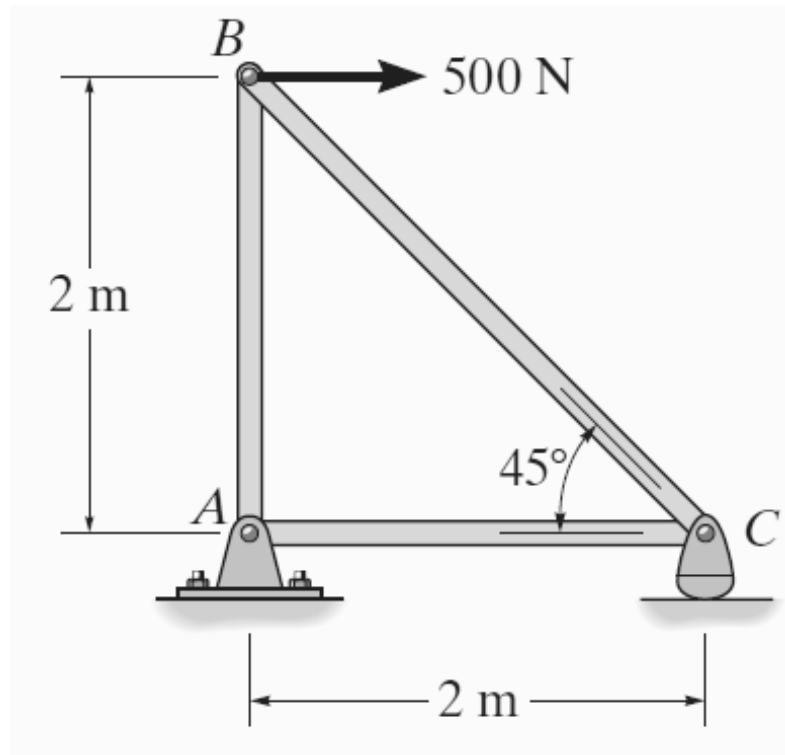


The forces in the members of this simple roof truss can be determined using the method of joints.

Concluding Notes:

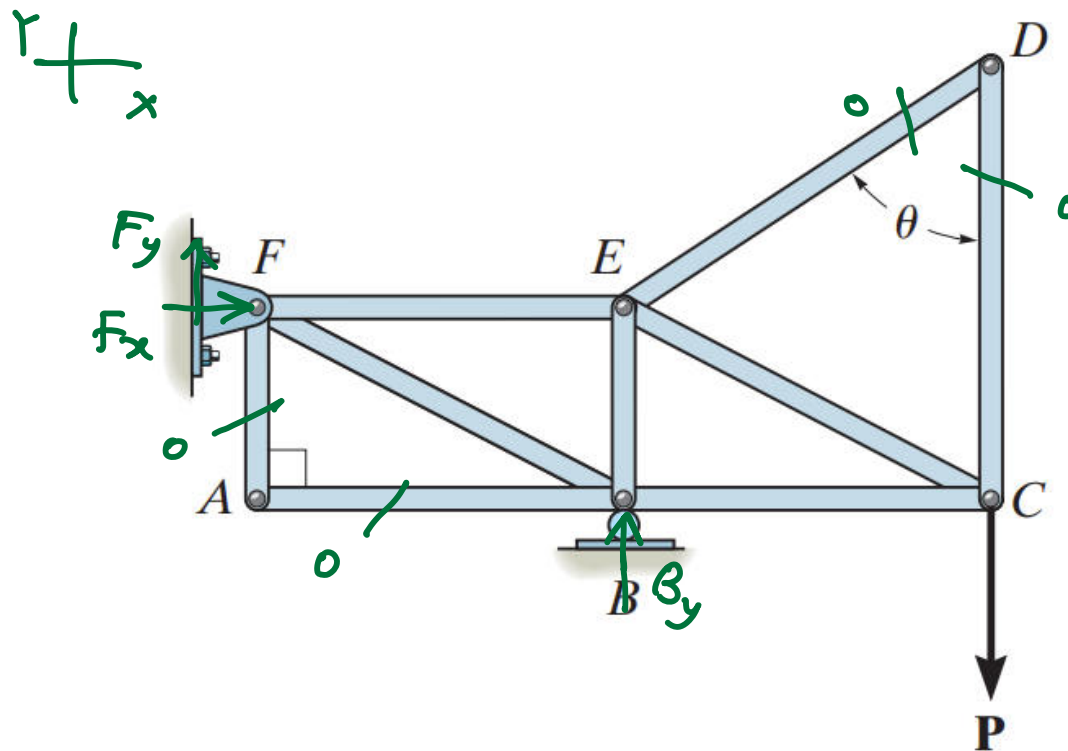
- Simple trusses are composed of triangular elements. The members are assumed to be pin connected at their ends and the loads applied at the joints.
- If a truss is in equilibrium, then each of its joints is in equilibrium. The internal forces in the members become external forces when the free-body diagram of each joint of the truss is drawn. A force pulling on a joint is caused by tension in a member, and a force pushing on a joint is caused by compression.

Example: Determine the force in each member of the truss shown below and indicate whether the members are in tension or compression.



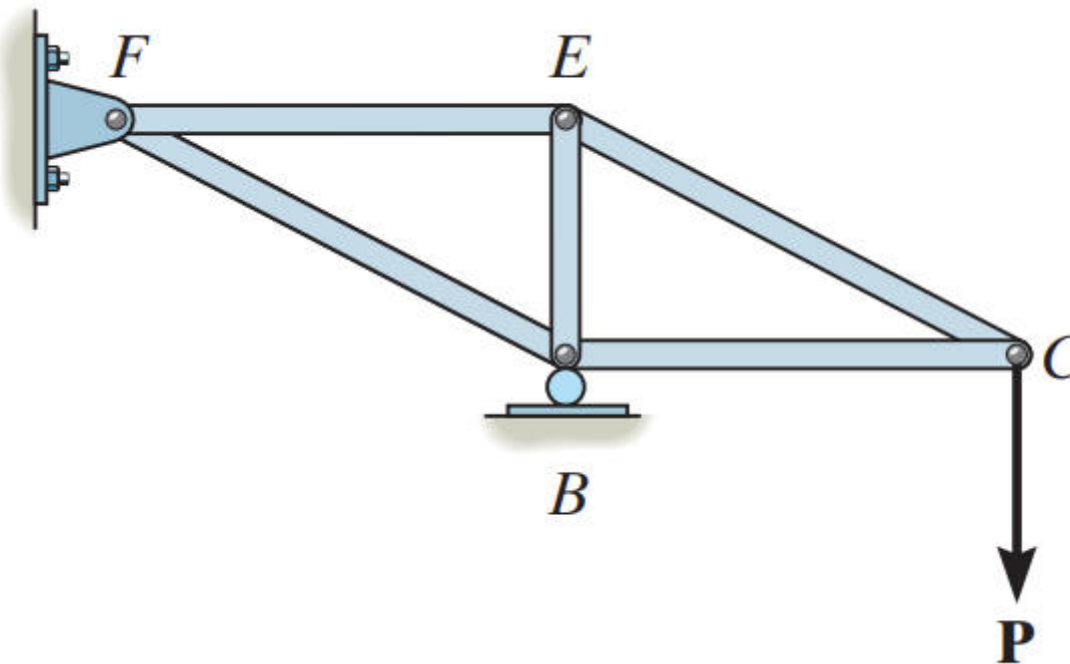
Methods of Analysis: Zero-Force Members

Truss analysis using the method of joints is greatly simplified if we can first identify those members which support *no loading*. These zero-force members are used to increase the stability of the truss during construction and to provide added support if the loading is changed.



Methods of Analysis: Zero-Force Members

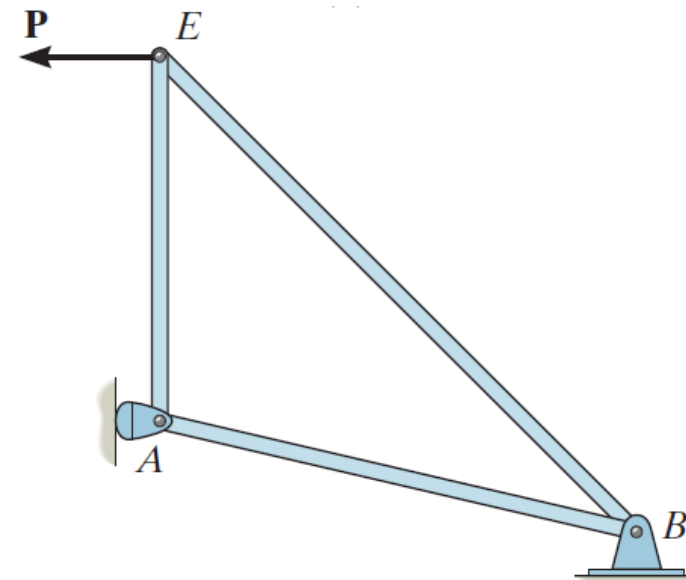
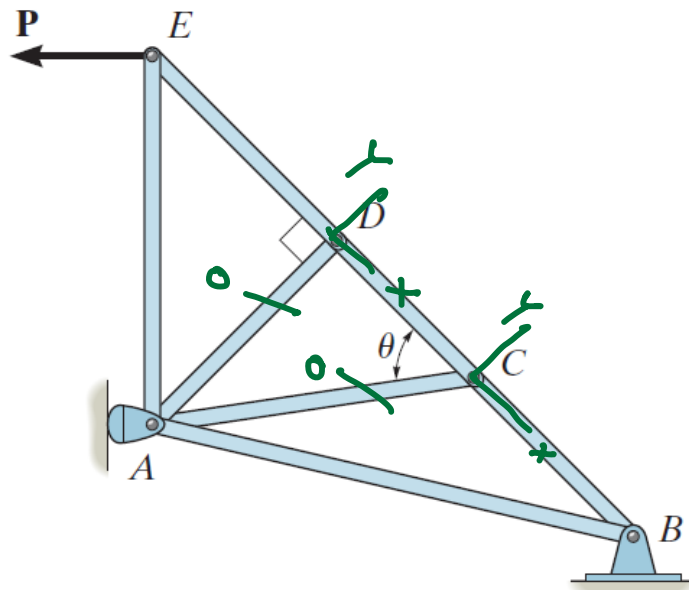
Truss analysis using the method of joints is greatly simplified if we can first identify those members which support *no loading*. These zero-force members are used to increase the stability of the truss during construction and to provide added support if the loading is changed.



Methods of Analysis: Zero-Force Members

To summarize, then, *if only two non-collinear members form a truss joint and no external load or support reaction is applied to the joint, the two members must be zero-force members.*

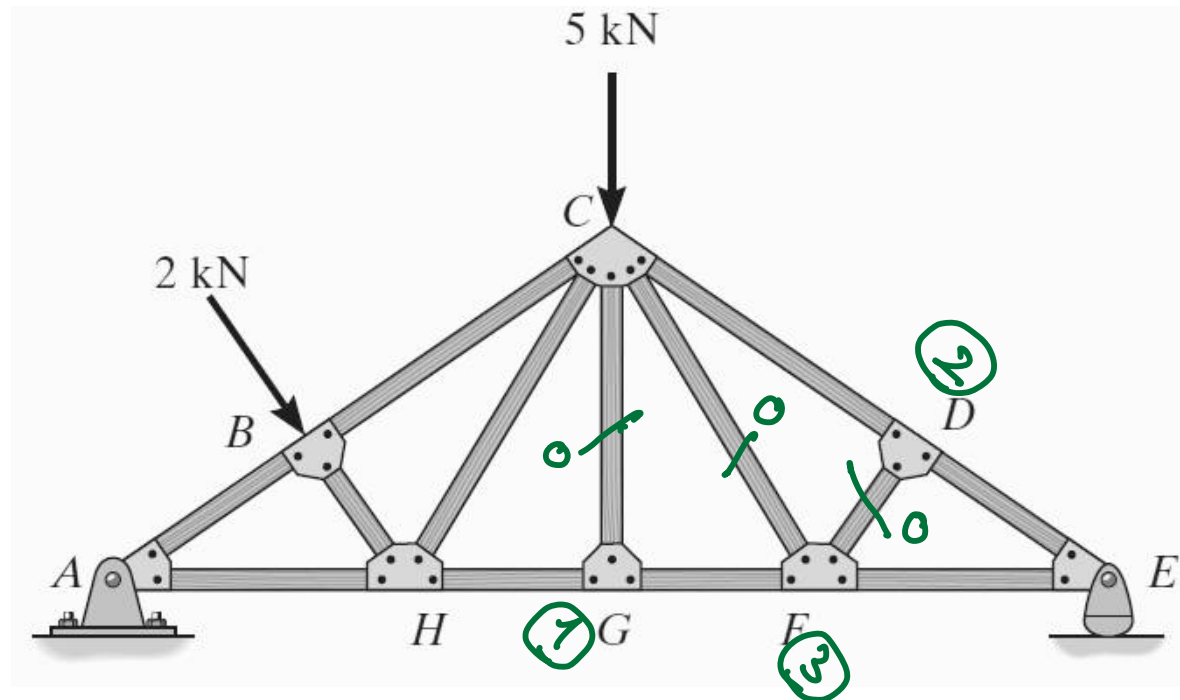
If three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction has a component that acts along this member.



Concluding Notes:

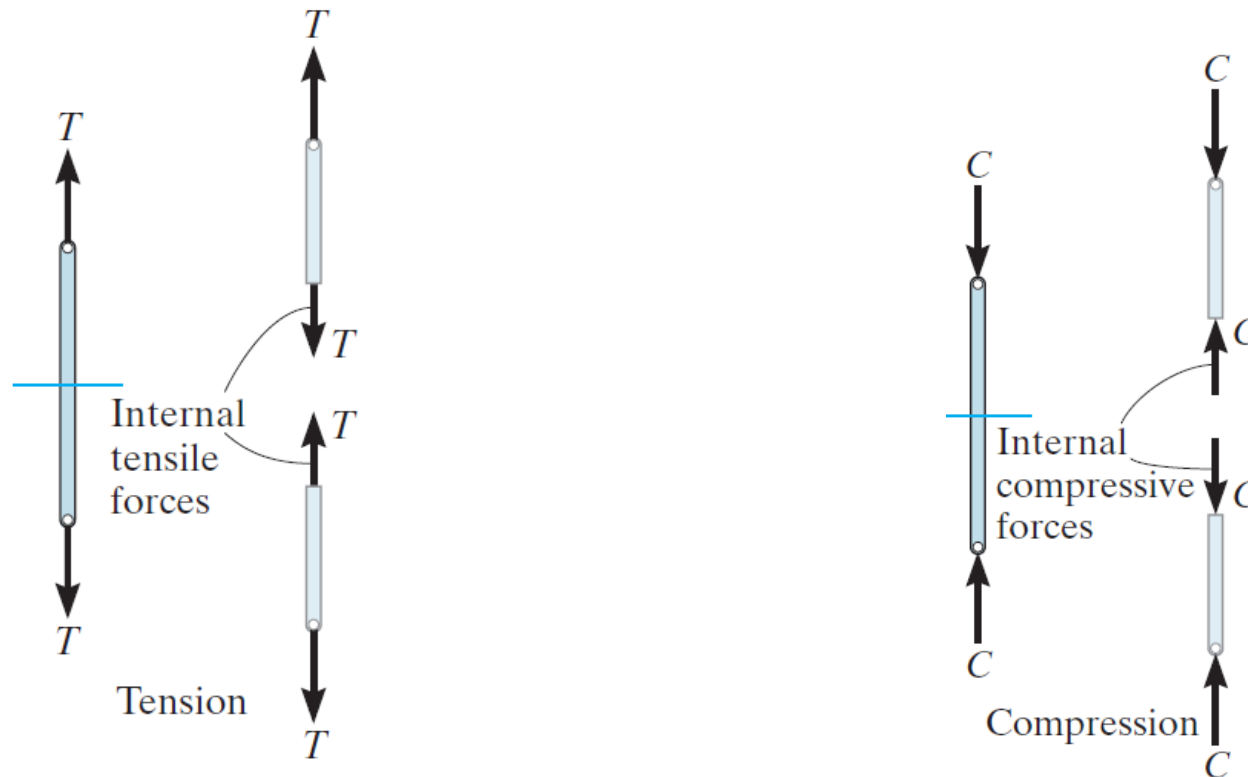
- Zero-force members support no load; however, they are necessary for stability, and are available when additional loadings are applied to the joints of the truss. These members can usually be identified by inspection. They occur at joints where only two members are connected and no external load acts along either member. Also, at joints having two collinear members, a third member will be a zero-force member if no external force components act along this member.

Example: Using the method of joints, determine all the zero-force members of the *Fink roof truss* shown below. Assume all joints are pin connected.



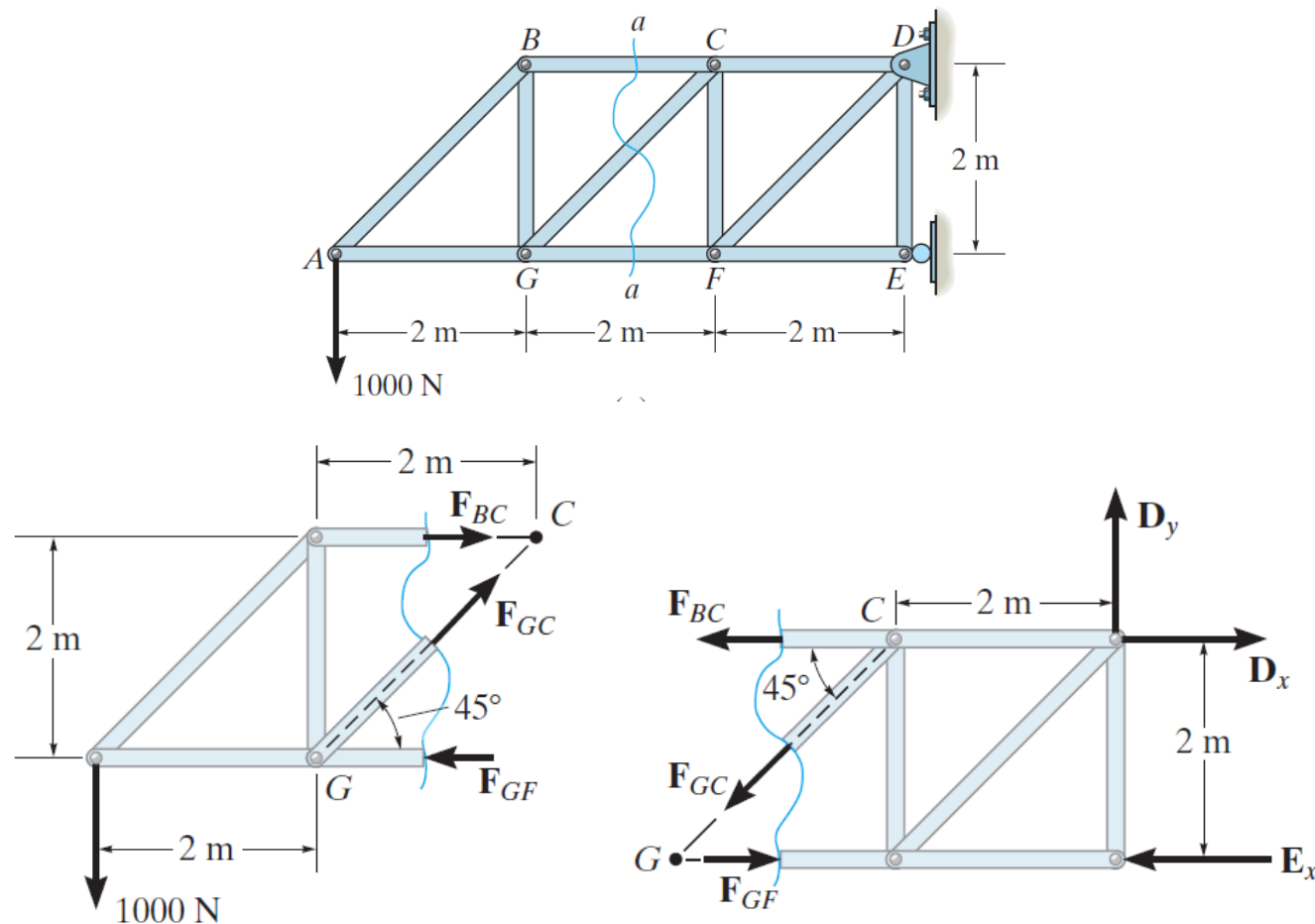
Methods of Analysis: The Method of Sections

When we need to find the force in only a few members of a truss, we can analyze the truss using the *method of sections*.



Methods of Analysis: The Method of Sections

The method of sections can also be used to “cut” or section the members of an entire truss.

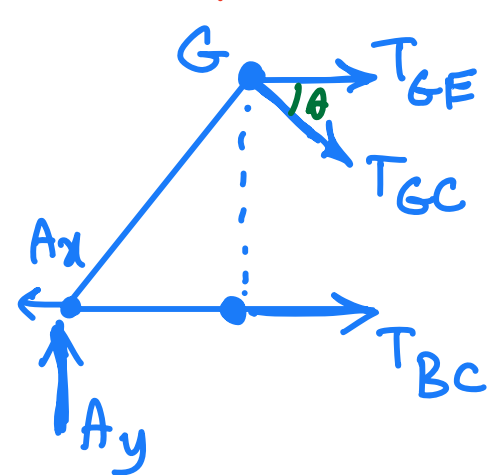


Concluding Notes

- If a truss is in equilibrium, then each of its parts is in equilibrium. The internal forces in the members become external forces when the free-body diagram of a part of the truss is drawn. A force pulling on a member causes tension in the member, and a force pushing on a member causes compression.

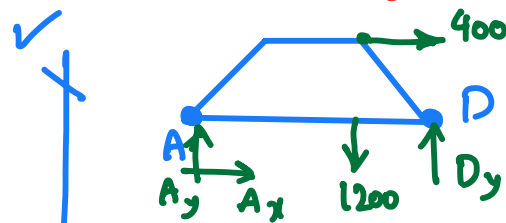
Example: Determine the force in members GE , GC , and BC of the truss shown below. Indicate whether the members are in tension or compression.

Method of sections:



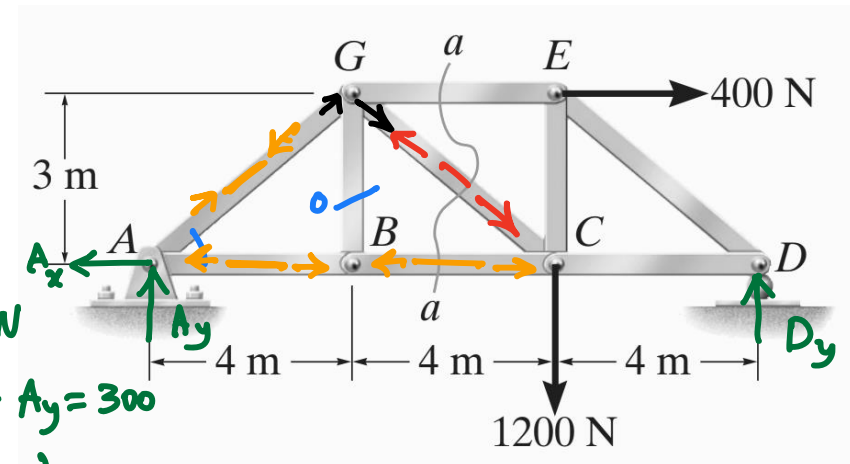
$$\begin{aligned}\sum F_x &= T_{GE} + T_{BC} - A_x \\ &\quad + T_{GC} \cos \theta = 0 \\ \sum F_y &= -T_{GC} \sin \theta + A_y = 0 \\ \sum M_G &= T_{BC}(3) - A_y(4) - A_x(3) = 0\end{aligned}$$

Method of joints:



$$\begin{aligned}\sum F_x &= A_x + 400 = 0 \rightarrow A_x = -400 \text{ N} \\ \sum F_y &= A_y + D_y - 1200 = 0 \rightarrow A_y = 300 \text{ N} \\ \sum M_A &= -8(1200) + 12(D_y) - 3(400) = 0 \\ D_y &= 900 \text{ N}\end{aligned}$$

$$\begin{aligned}\sum F_x &= T_{AB} + T_{AG} \left(\frac{4}{5}\right) - 400 = 0 \\ \sum F_y &= 300 + T_{AG} \left(\frac{3}{5}\right) = 0 \\ T_{AG} &= -500 \text{ N}, T_{AB} = 800 \text{ N} \\ \text{AG: Compression} \quad \text{AB: Tension}\end{aligned}$$

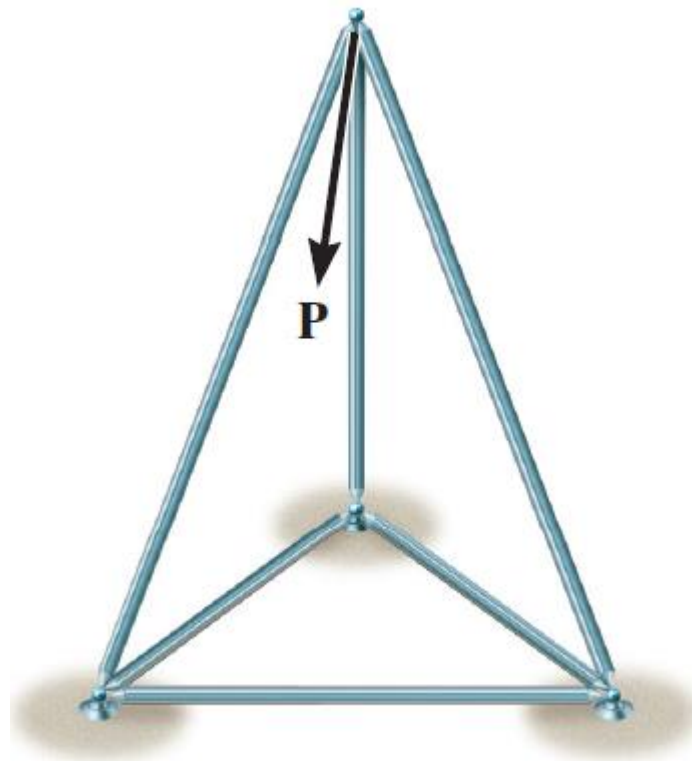


$$\begin{aligned}\sum F_x &= T_{AB} + T_{BC} - 400 = 0 \\ T_{BC} &= T_{AB} = 800 \text{ N} \\ \text{BC: Tension}\end{aligned}$$

$$\begin{aligned}\sum F_x &= T_{GE} + T_{GC} \left(\frac{4}{5}\right) + T_{AG} \left(\frac{4}{5}\right) = 0 \\ \sum F_y &= T_{AG} \left(\frac{3}{5}\right) - T_{GC} \left(\frac{3}{5}\right) = 0 \\ T_{GC} &= 500 \text{ N tension}\end{aligned}$$

Space Trusses

A ***space truss*** consists of members joined together at their ends to form a stable three-dimensional structure.



A **tetrahedron** space truss.

Space Trusses



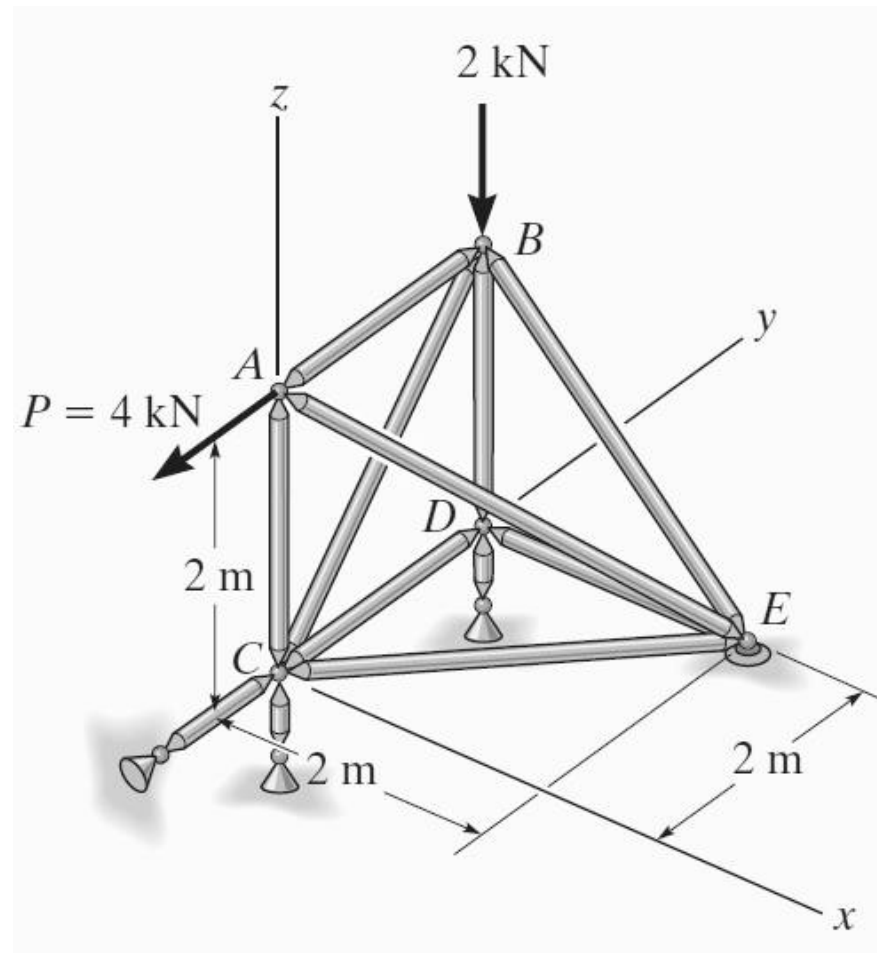
Typical roof-supporting space truss. Notice the use of ball-and-socket joints for the connections.

Space Trusses



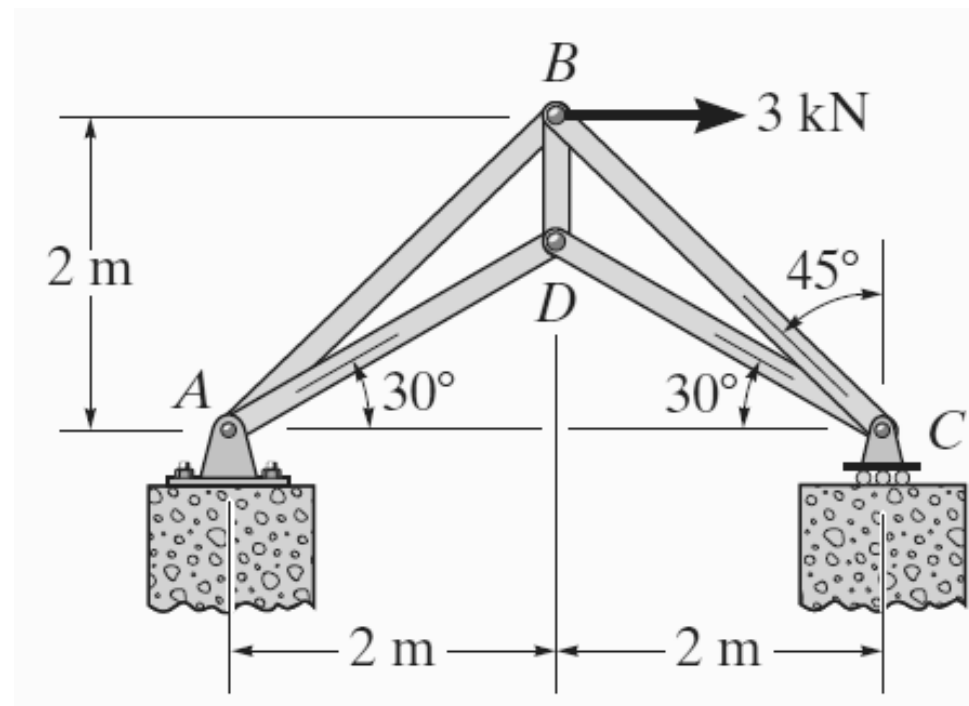
For economic reasons, large electrical transmission towers are often constructed using space trusses.

Example: Determine the forces acting in the members of the space truss shown below. Indicate whether the members are in tension or compression.



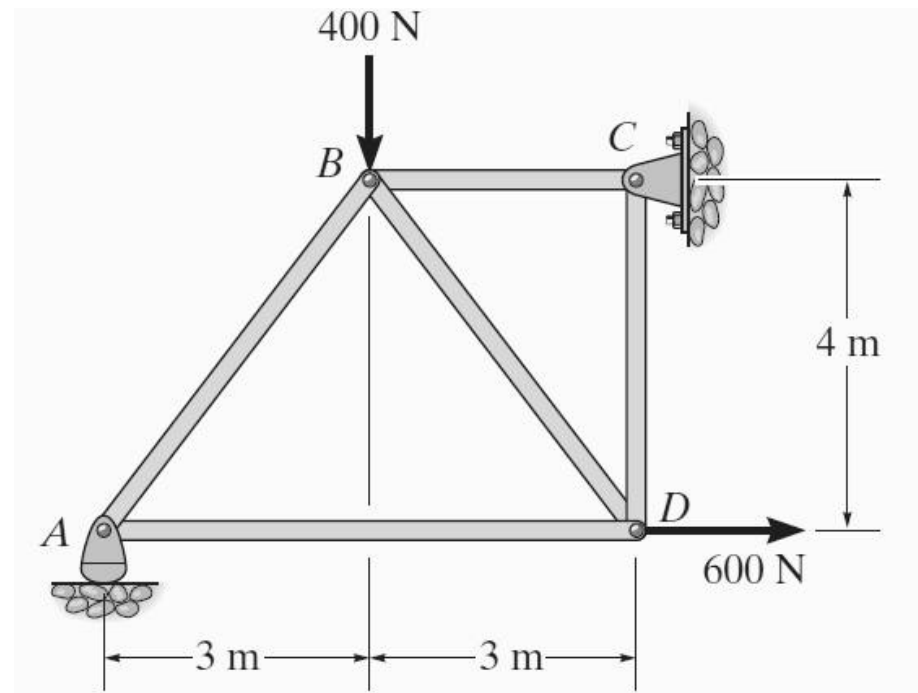
Practice at Home

1. Determine the forces acting in all the members of the truss shown below and indicate whether the members are in tension or compression.



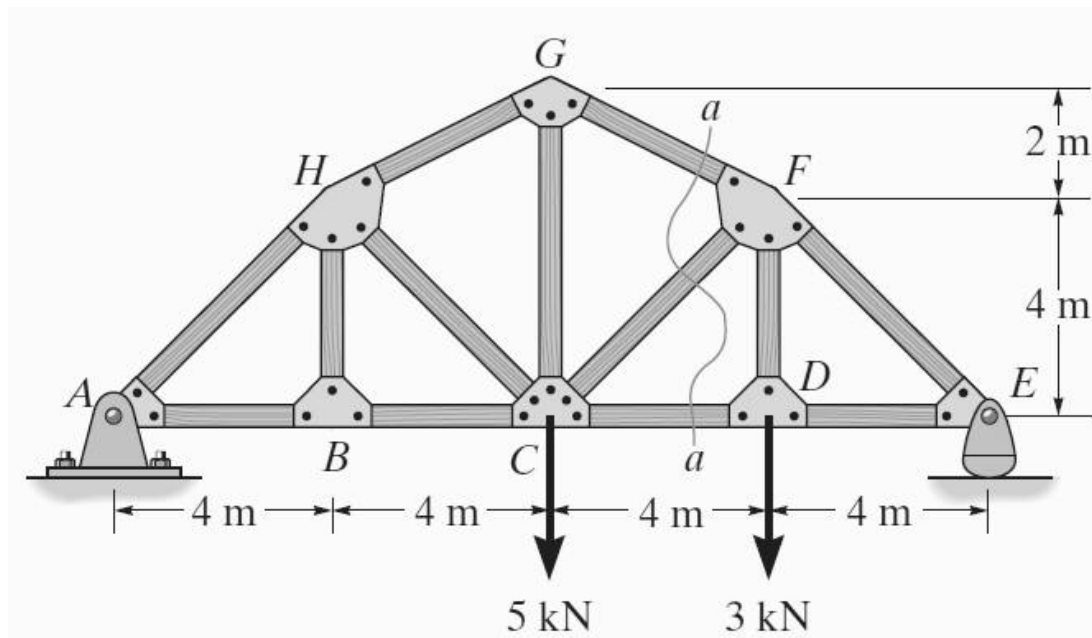
Practice at Home

2. Determine the force in each member of the truss shown below and indicate whether the members are in tension or compression.



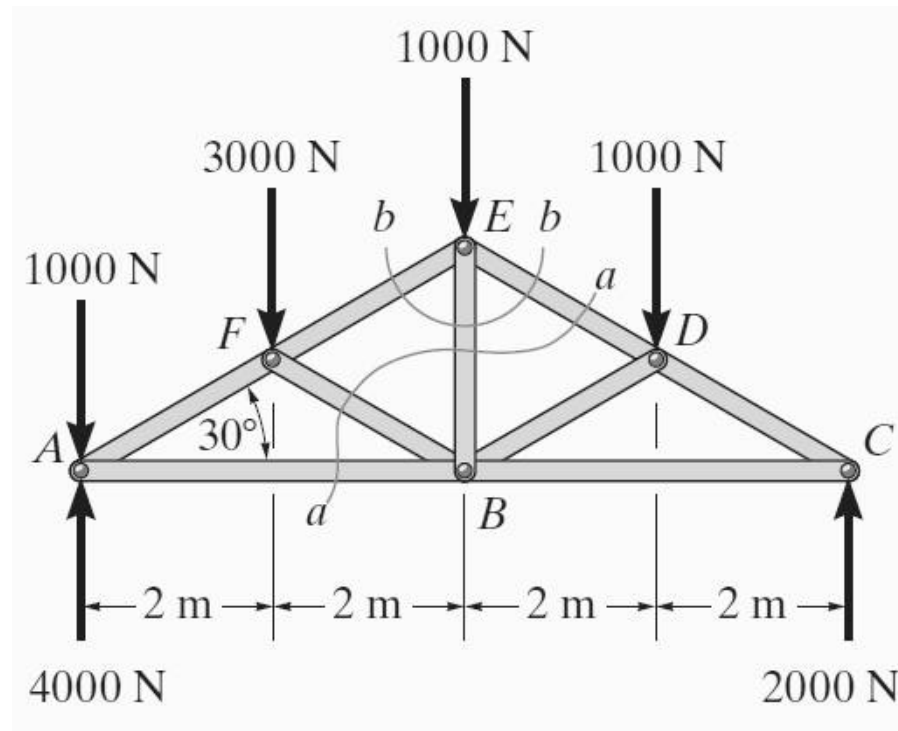
Practice at Home

3. Determine the force in member CF of the truss shown below. Indicate whether the member is in tension or compression. Assume each member is pin connected.



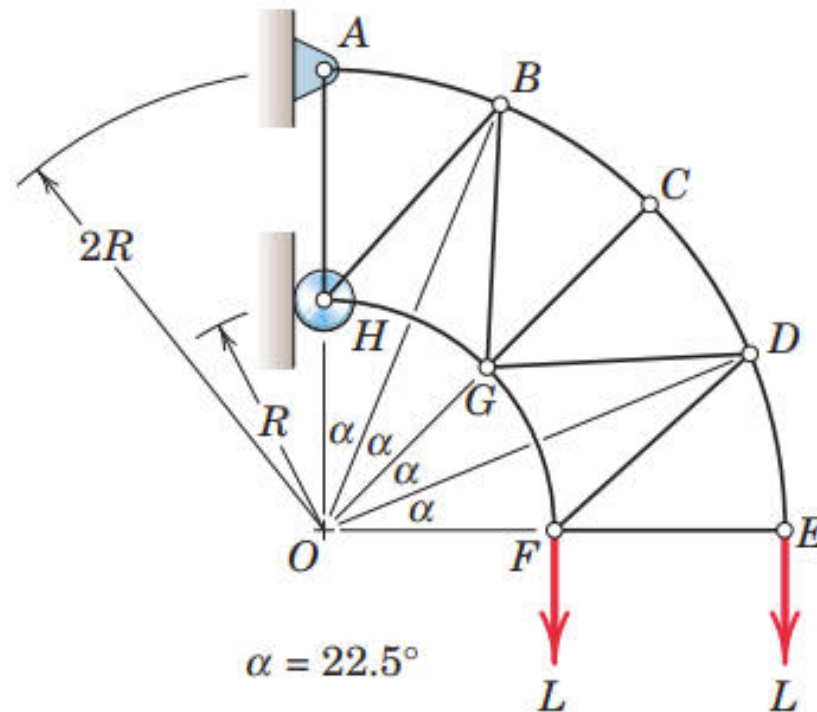
Practice at Home

4. Determine the force in member EB of the roof truss shown below. Indicate whether the member is in tension or compression.



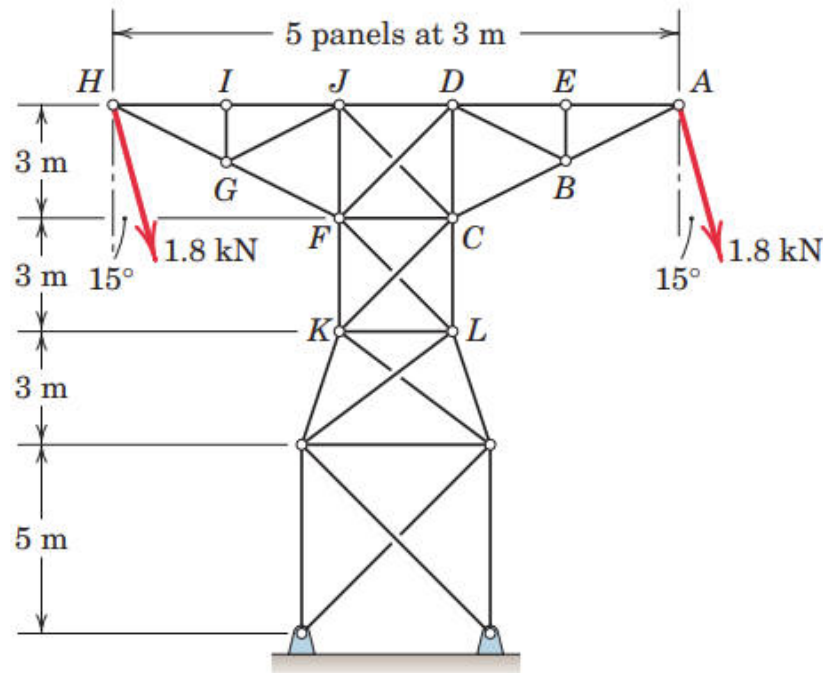
Practice at Home

5. The simple truss shown supports the two loads, each of magnitude L . Determine the forces in members DE , DF , DG , and CD .



Practice at Home

6. The tower for a transmission line is modeled by the truss shown. The crossed members in the center sections of the truss may be assumed to be capable of supporting tension only. For the loads of 1.8 kN applied in the vertical plane, compute the forces induced in members AB , DB , and CD .



Practice at Home

7. Determine the force in members EF , AF , and DF of the space truss and state if the members are in tension or compression. The truss is supported by short links at A , B , D , and E .

